

1. Regression (general idea)

Regression is a method in statistics used to **predict a value (number)** based on other variables.

Example: Predicting house price based on size, location, etc.

2. Linear Regression

A type of regression that assumes a **straight-line relationship** between input and output.

It tries to fit a line like:

👉 $y = mx + c$

Example: Predicting salary based on years of experience.

3. Logistic Regression

Despite the name, it's used for **classification (yes/no outcomes)**, not continuous values.

It uses an S-shaped curve (sigmoid) to output probabilities between 0 and 1.

Example: Predicting whether an email is spam or not.

In one line:

- Regression → Predict numbers
- Linear Regression → Straight-line prediction
- Logistic Regression → Predict categories (usually binary)

Feature	Linear Regression	Logistic Regression
Purpose	Predict continuous values	Predict categorical outcomes (usually binary)
Output	Any real number (e.g., 25, 100.5)	Probability (0 to 1), then classified (e.g., Yes/No)
Example	Predicting house price	Predicting spam vs not spam
Model Equation	$y = mx + c$	Uses sigmoid function: $1 / (1 + e^{-x})$
Graph Shape	Straight line	S-shaped (sigmoid curve)
Type of Problem	Regression problem	Classification problem
Output Range	$(-\infty, +\infty)$	$(0, 1)$
Interpretation	Direct numeric prediction	Probability of belonging to a class
Error Measurement	Mean Squared Error (MSE)	Log Loss (Cross-Entropy)
Algorithm Type	Based on statistics	Based on statistics